

On the Opportunities and Risks of Foundation Models for Natural Language Processing in Radiology

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Automated text analysis—also known as natural language processing or NLP—has many potential applications in radiology, including information extraction from radiology reports or other medical text, text classification, text summarization, and text generation (1). Modern techniques for NLP employ large artificial neural networks—most often, derivatives of the transformer architecture (2)—that require pretraining on vast corpora of text data (on the order of millions of documents) to achieve state-of-the-art performance on text prediction tasks. Such large language models are referred to as *foundation models* because they serve as the foundation to train models for more specific tasks, such as sentiment classification or text summarization. The application of such foundation models is referred to as *transfer learning*. Some of the foundation model's general semantic knowledge is transferred to the more specific task via a process called *fine-tuning*, whereby incremental adjustments to the model are learned from a typically much smaller quantity of task-specific data.

Bidirectional encoder representations from transformers (BERT) is a large language model that leverages the transformer architecture to encode the context of “tokens” (often words or word pieces) on either side of a target token in a stream of text (eg, a sentence in a document) (3). The methods by which the original BERT model was trained make it particularly good at predicting missing (or “masked”) tokens and extracting information from text. While the BERT model is a general large language model, it may be fine-tuned on domain-specific corpora, which may result in improved performance on domain-specific tasks, such as analysis of biomedical text. Variants such as ClinicalBERT have been fine-tuned on large corpora of medical text, including de-identified clinical notes (4). These models can then be further fine-tuned on relatively smaller corpora of text data such as radiology reports for narrower tasks, such as classifying a radiology report as “normal” or “abnormal” (5).

Though transformer-based models pretrained on radiology reports have been previously reported, these models were trained on relatively small datasets with a limited range of modalities and anatomic coverage. For instance, RadBERT-CL, an algorithm developed for multilabel classification, resulted from fine-tuning BlueBERT on the MIMIC-CXR dataset of 377 110 chest radiography reports (6). BI-RADS BERT, a tool designed to divide a report into standard sections (eg, “Findings”, “Impression”) and extract information from specific fields, was developed by fine-tuning BERT on 155 000 breast imaging reports, including mammography, US, MRI, and biopsy procedure reports (7). Although researchers may find a number of uses for these models and tools, there remains great potential for foundation models trained on very large datasets of radiology reports from multiple institutions and comprising a diverse range of modalities and anatomic locations to serve as the backbone for transfer learning tasks for radiology use cases.

Foundation models can facilitate NLP research and model development at all scales. The process of training such a model involves the encoding of semantic information about individual words or tokens into high-dimensional embeddings. This process is highly dependent on the context in which the words are used. Consider the lay use of the words *negative* and *consolidation* in contradistinction to their use in the medical literature. Now consider the general scientific use

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See also the article by Yan et al in this issue.

Conflicts of interest are listed at the end of this article.

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of the modifiers *indistinct* and *ill-defined* as compared with their use in the context of a mammography report describing a breast mass. It should become clear that the semantic information changes quite a bit when transitioning between lay, scientific, clinical, and radiologic dialects of the English language. Encoding rich semantic information requires vast numbers of examples in a variety of contexts. However, as we drill down into more specific dialects, the general semantics of a word become less useful. For this reason, fine-tuning an NLP model that was pretrained on a more specific corpus of text data is more likely to generate a more useful and generalizable model, particularly when the amount of training data available for the narrower task is small. That said, it takes a large amount of heterogeneous data for a model to learn the nuanced semantics of a given dialect.

In this issue of *Radiology: Artificial Intelligence*, Yan et al (8) report the results of experiments in pretraining various general, scientific, and clinical variants of the BERT architecture on a large corpus of radiology reports (4.42 million reports) from the U.S. Department of Veterans Affairs health care system to create six “RadBERT” models. Each RadBERT model and its parent model were tested on three tasks: abnormal sentence classification, report coding, and report summarization. In keeping with the expectations of the above line of logic, the model that performed the best on these tasks was derived from one of the language models ontologically most similar to the “radiology dialect”: BioMed-RoBERTa. The authors’ analysis showed that the performance of the RadBERT models outperformed the respective baselines set by their parent models, underscoring the significance of fine-tuning clinical language models on radiology text prior to their application to radiologic use cases.

The ability to fine-tune foundation models for more specific dialects offers an alternative to the training of transformer-based models de novo in resource-constrained contexts. Training large language models using millions of radiology reports is not a practical approach for individual institutions. Most research groups and practices hoping to leverage transformer-based models for radiologic use cases will need access to large language models to facilitate transfer learning on local, task-specific data. In their experiments, Yan et al (8) showed that one of the RadBERT models fine-tuned on 5% of available training data performed similarly to the BioBERT model fine-tuned on 100% of the data. This result provides strong support for the relative potency of a domain-specific pretrained model compared with more general large language models. Accordingly, access to the RadBERT models may enable the development of robust task-specific models with a lower data requirement. Furthermore, the ability to fine-tune pretrained, domain-specific models offers a significant reduction in the cost, time, and environmental impact associated with developing large language models. For example, BioBERT required pretraining for 23 days on eight industry-grade GPUs, and RoBERTa required pretraining for 1 day on 1024 such GPUs (9). The estimated carbon footprint of those training regimes is roughly equivalent to burning between 66 and 286 kg of coal (<https://mlco2.github.io/impact/#co2eq>). Furthermore,

the cost of such training is likely not feasible for most groups without substantial external funding. However, fine-tuning the RadBERT models should require significantly smaller training datasets and less computational time, thus making such development more attainable for smaller outfits and with less environmental impact.

The training and sharing of the RadBERT models—as the authors have stated they will do—represents an important step in radiology NLP research and model development. These models will be the first radiology-specific language models using a transformer-based architecture and pretrained on a comprehensive radiology report corpus of this scale. However, as the ancient adage goes, “With great power comes great responsibility.” Foundation models entail potential and demonstrated risks of bias, inequity, and misuse, among other concerns (10). Any biases in the underlying training data will likely be encoded in the model and its associated embeddings, which will thus likely be inherited by any models fine-tuned from the RadBERT models. As more tools are developed from foundation models for clinical use, infrastructure and processes for monitoring performance over time will be required to limit inequities in patient care that may result from these biases. A detailed discussion of these risks and strategies for mitigating them is beyond the scope of this commentary. For further reading on this subject more generally, please see the referenced manuscript by Bommasani et al (10), which was a source of inspiration for the title and the content of this writing.

Given the challenges of de-identifying and sharing radiology report data broadly, the establishment of radiology-specific foundation models on large corpora of radiology reports spanning a range of subspecialty areas is a timely benchmark for radiology NLP research and tool development. Further work on this frontier will reveal limitations and ethical dilemmas that will need to be addressed with appropriate care and caution. However, it is our belief that the RadBERT models will accelerate the development and usage of transformer-based NLP models, helping the field realize substantial efficiency gains in imaging workflows and radiology research.

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References

1. Linna N, Kahn CE Jr. Applications of natural language processing in radiology: A systematic review. *Int J Med Inform* 2022;163:104779.
2. Vaswani A, Shazeer NM, Parmar N, et al. Attention is all you need. arXiv:1706.03762 [preprint] <https://arxiv.org/abs/1706.03762>. Posted June 12, 2017. Accessed June 15, 2022.
3. Devlin J, Chang M, Lee K, Toutanova K. BERT: Pre-training of deep bidirectional transformers for language understanding. arXiv:1810.04805 [preprint] <https://arxiv.org/abs/1810.04805>. Posted October 11, 2018. Accessed June 15, 2022.
4. Huang K, Altosaar J, Ranganath R. ClinicalBERT: Modeling Clinical notes and predicting hospital readmission. arXiv:1904.05342 [preprint] <https://arxiv.org/abs/1904.05342>. Posted April 10, 2019. Accessed June 15, 2022.

5. Wiggins WF, Kitamura F, Santos I, Prevedello LM. Natural language processing of radiology text reports: interactive text classification. *Radiol Artif Intell* 2021;3(4):e210035.
6. Jaiswal A, Tang L, Ghosh M, Rousseau JF, Peng Y, Ding Y. RadBERT-CL: Factually-aware contrastive learning for radiology report classification. *Proc Mach Learn Res* 2021;158:196–208.
7. Kuling G, Curpen B, Martel AL. BI-RADS BERT and using section segmentation to understand radiology reports. *J Imaging* 2022;8(5):131.
8. Yan A, McAuley J, Lu X, et al. RadBERT: Adapting transformer-based language models to radiology. *Radiol Artif Intell* 2022;4(4):e210258.
9. Liu Y, Ott M, Goyal N, et al. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv:1907.11692* [preprint] <https://arxiv.org/abs/1907.11692>. Posted July 26, 2019. Accessed June 15, 2022.
10. Bommasani R, Hudson DA, Adeli E, et al. On the opportunities and risks of foundation models. *arXiv:2108.07258* [preprint] <https://arxiv.org/abs/2108.07258>. Posted August 16, 2021. Accessed June 15, 2022.